PRAKRITI
The Conservation club
IISER Tirupati.

INTERVIEW – 1 (C)
SUSTAINABLE ARCHITECHTURE

On 30th May, Prakriti Club (The conservation club of IISER-Tirupati) had the opportunity of conducting a written interview with **Ar. Aditya,** working in design studio, Delhi**,** India.

The written interview questions were prepared by Nithish.G.S (Prakriti club member) and the interview was conducted by Anchita Sharma (Prakriti club member) and through the answers, we understood the responsibility of green buildings in making the future more earth friendly! The interview:

What are current trends in sustainable architecture and green building projects amongst the architects?

GREEN BUILDING AND NET ZERO CONSTRUCTION - The energy efficiency of a building involves considering the energy used by the building before, during, and after construction, which includes building services such as air conditioning, lighting, etc. Net-zero energy refers to the annual amount of energy created on-site equal to the total energy consumed by the building. Various rating systems like LEED help provide a framework to build green buildings all over the world.

CIRCULAR ECONOMY FOR SUSTAINABILITY - Circular economy refers to Reduce-Reuse-Recycle. Architecture that involves this system aids in reducing wastage generated during construction or materials used on an everyday basis. Several architects also make use of plastic, glass bottles, etc. to cut down on wastage.

SMALL SCALE CONSTRUCTION - The significance of large mansions and structures are decreasing due to land shortage and high-density areas. A house has come down to its basic necessity with more importance given to its functionality and open spaces. With foldable and inbuilt furniture, an entire house can fit in one room area.

BIOPHILIC ARCHITECTURE - Biophilic architecture is a sustainable approach that incorporates nature into a structure. Such structures can act as a breathing lung in a metropolitan city or blend with the existing landscape of the place. Roof gardens and green walls are a few elements used that can incorporate vegetation with minimal space used.

ADAPTIVE REUSE - Adaptive reuse refers to modifying an existing building to present needs and remodeled with the addition of new functions. Reuse of several historic buildings is a great way to bring purpose in a building, which otherwise becomes deteriorated or left unused. It can even bring a new character to the existing structure and deliver an alternate meaning without compromising on its cultural significance.

How developing countries like India are foreseeing the possibility of sustainable architecture and green buildings?

One challenge to India's acceptance of sustainable architecture is the gap between technology and economic status. On one end,

sophisticated technology-based solutions have been developed to improve the energy efficiency of buildings, but they require a high initial investment that very few can afford. On the other end, affordable, low-cost technologies, such as mud architecture, are already available; however, these do not fit in with the aspirations of the upwardly mobile urban population. Affordable technology-based solutions are thus seen as the only means of addressing environmental degradation.

Which world countries are encouraging sustainable designs and welcoming innovative architecture projects (from the current generation architect community's point of view)?

The Leadership in Environmental and Energy Design or LEED, certification has become the world's most widely used and recognized green building rating system. The U.S. Green Building Council (USGBC), which sets LEED certification standards, announced last week its international rankings of the Top 10 Countries for LEED outside of the U.S. U.S. being the pioneer of sustainable design and LEED tops the list with most area of green building projects.



Any internship experiences and innovative sustainable designs from your student projects which you would like to illustrate to environmentalists and climate activists?

The most prominent ways which I learned from my internship is to save energy which can be achieved by reducing the heat gain and reducing the HVAC (Heating, Ventilation, and Air Conditioning) load on the building. This can be done by a lot of ways liked using passive building design, high performance glass and building envelope.

The awareness about green buildings in metropolitan cities (from an aspiring architect's point of view)?

Many green building consultancies have emerged which are working on the green building design and many architects claim to work on sustainable design in most of the metropolitan cities like Delhi, Mumbai, Ahmedabad, Chandigarh, Bangalore etc. and they are helping in a way to create awareness among the clients. Though it would still be a relevant conclusion that sustainability should come in more simple ways

which are cohesive with the nature and not just numbers and material technology.

Your personal inspiration when it comes to sustainable design and his/her major contribution?

My personal inspiration comes from the idea of "Don't disturb what you don't have to". While construction it is obvious that the current situation of the site will be altered but it's a smart idea to not disturb nature when you can work along with it and also finding alternative ways to add back to nature which we have taken. This can also include use of traditional materials in an innovative way. Adaptive reuse of materials is also a great way to add to the sustainability of a project.

Your final comments about the future of sustainable architecture and green buildings?

Consumer attitude towards environment has a great impact towards adapting green buildings. Keeping in mind the novelties, consumers are also inclined towards this sustainable change. Over the recent years growing awareness about the benefits of using eco-friendly items, has prompted a positive and pro-active response from manufacturers as well as developers across the country. However, the permeability of this phenomenon has to be still instilled amongst the consumers. Despite the acceptance of the concept of sustainable living, widespread adoption of the same will require time, considering the present volatile growth graph of the real estate sector, which may pose as a greater challenge for many to invest in. However, irrespective of such bottlenecks, no one can deny that Green Buildings are the need of the hour. It might appear as a comparatively expensive investment, but in the long run, the returns both tangible and intangible are much higher, in comparison to any regular structure. The low operational costs, reduced energy expense and carbon emission loads are a definite upside to such projects. Thus, it holds a better prospect in terms of higher resale or rental values. In contemporary times where India is heading towards shared economic spaces to amplify cost-efficiency, productivity and sustainability, it would be imperative to opt for an eco-friendlier lifestyle to ensure a better and brighter future for mankind.

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